Bluestock MF Capstone — Mutual Fund Analytics

Final Report | Bluestock Fintech Internship 2026

Author: Giri Gnana Sunder

Organization: Bluestock Fintech

Date: June 2026

Table of Contents

Executive Summary

This project delivers a full-stack mutual fund analytics pipeline covering 40+ Indian MF schemes from 2022 to 2026. Built during the Bluestock Fintech Data Analyst Internship, the pipeline ingests live NAV data, cleans and stores it in a relational database, performs exploratory and performance analytics, and presents findings through an interactive Power BI dashboard.

Key Outcomes

- **46,000+** NAV records ingested from mfapi.in
- **37,619** schemes catalogued from AMFI India
- **32,778** investor transactions analysed
- **15+** EDA charts generated
- **Fund Scorecard** ranking 40 funds on composite metrics
- **4-page** interactive Power BI dashboard built
- **Fund recommender system** configured for 3 distinct risk profiles

Data & ETL Pipeline

Data Sources

- **mfapi.in:** Live NAV history for 40 specific schemes.
- **AMFI India:** Broad fund master directory capturing 37,619 schemes.
- **Bluestock Datasets:** 10 comprehensive CSV files tracking metrics like AUM, SIP, ledger transactions, portfolio holdings, and benchmark indices.

ETL Process

> **Ingestion:** data_ingestion.py fetches live data via REST API endpoints.
>
> **Cleaning:** data_cleaning.py handles algorithmic sweeps tracking nulls, duplicate entries, structural date formats, and NAV numeric validation.
>
> **Loading:** load_db.py pipelines database staging into relational SQLite models leveraging SQLAlchemy architectures.

Database Schema (5 Tables)

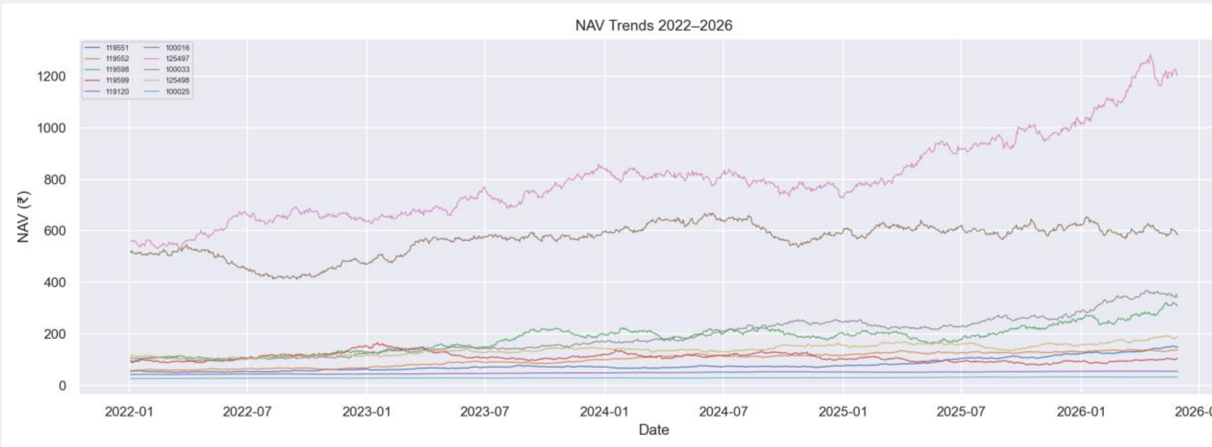
- **dim_fund** — Fund master data baseline reference dimension structure.
- **fact_nav** — High frequency time-series record tracking operational daily NAV movements.
- **fact_transactions** — Central ledger summarizing retail investor transaction blocks.
- **fact_performance** — Schema storing core analytical performance calculation outputs.
- **fact_sip_inflows** — System ledger aggregating tracked monthly SIP inbound funding lines.

Data Quality Profiles

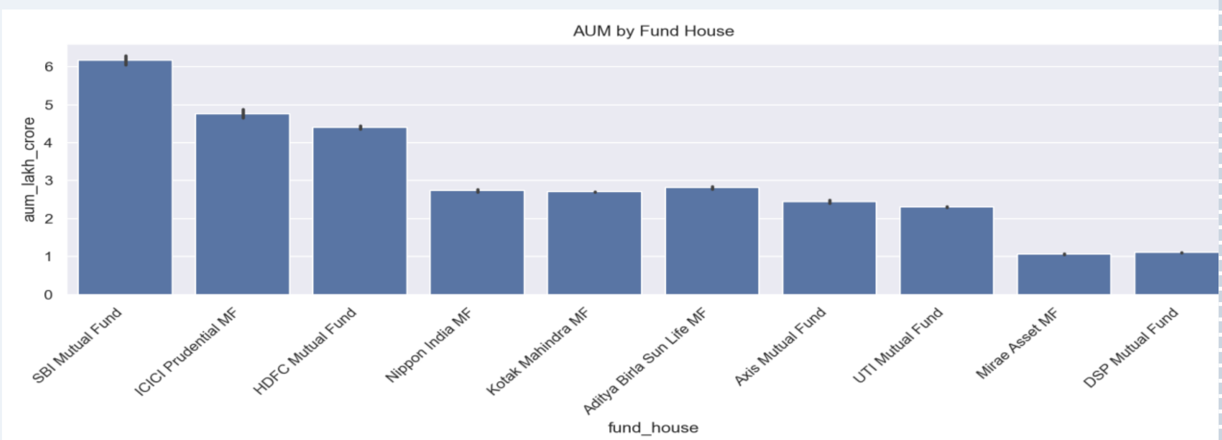
- 0 Missing or null asset value rows (NAV profiles).
- 0 Unmatched or orphaned AMFI scheme tracking codes.
- **Temporal Range:** January 2022 through May 2026 matrix footprint.

Exploratory Data Analysis Findings

[Visual Asset Container: chart1_nav_trends.png]



[Visual Asset Container: chart2_aum.png]



Core Exploratory Indicators (1-2)

1. SBI Mutual Fund asserts market volume control, dominating overall AUM matrix metrics scaling past 6+ Lakh Crore.

2. Aggregate systemic SIP inflows observed linear operational acceleration expanding 2.5x from January 2022 baseline up to December 2025.

Exploratory Data Analysis Findings (Cont.)

Core Exploratory Indicators (3-10)

3. Highly liquid category assets consistently secure the maximum localized asset volume tracking inflows.

4. The distribution properties of historical NAV markers reflect an extended right-skew trend pattern—placing the structural majority beneath the ₹200 threshold mark.

5. Direct equity category schemes display strong positive mathematical correlation coefficients ranging from 0.6 to 0.9.

6. Active, distinct systemic SIP accounts scale higher across successive annual tracking cycles.

7. ICICI Prudential and HDFC Mutual Fund capture adjacent market layers, establishing tight tier-1 competition directly matching SBI's footprint.

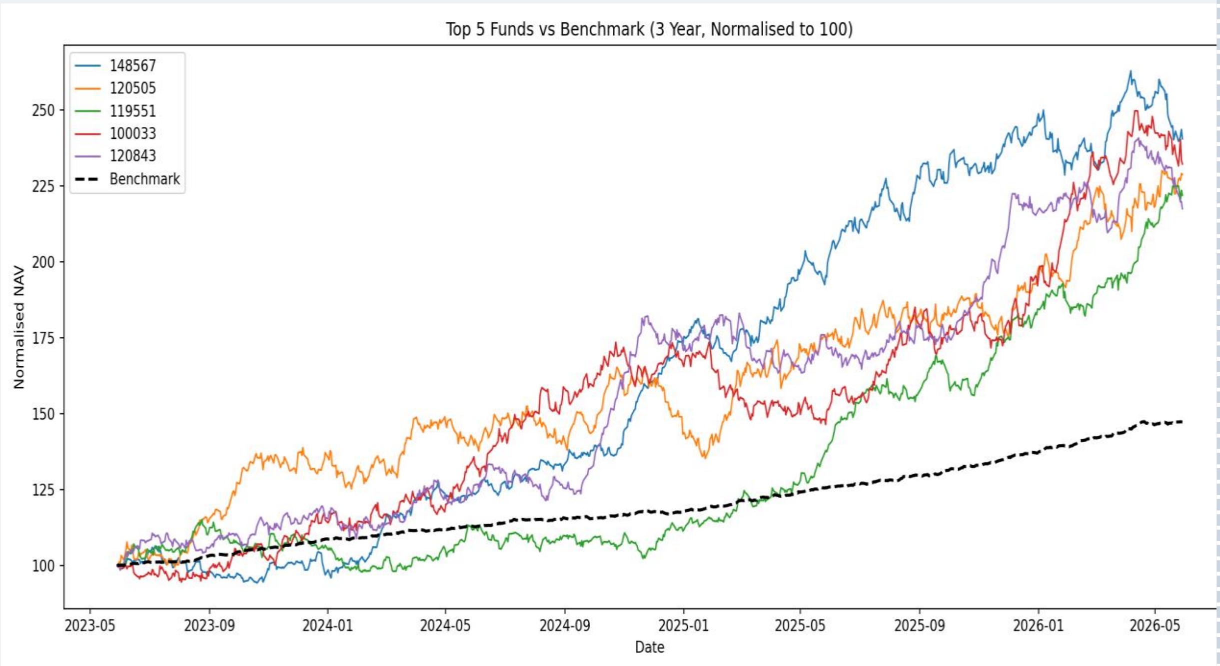
8. Highly specialized Sectoral/Thematic category funds display high inbound asset volume variance and volatile trends.

9. Year-over-Year (YoY) metrics for core system SIP growth factors reflect robust linear expansion without showing a single down-cycle phase.

10. Large Cap and Flexi Cap core allocations exhibit resilient, uniform asset growth profiles.

Performance Analysis

[Visual Asset Container: benchmark_chart.png]



Operational Engine Architecture Metrics

- **CAGR Calculation:** Annualized across standard 252-day business trading models. Peak performance ledger asset recorded a maximum 1-year trailing window loop hitting 50.7%.
- **Sharpe Ratio Evaluation:** Standardized via mathematical formula models. Base risk-free yield rate curves set flat at Rf = 6.5%. Top executing scheme marker generated an alpha efficiency score of 1.45.
- **Sortino Ratio Optimization:** Focuses calculations exclusively against down-side semantic trailing deviations to ensure downside volatility safety transparency for retail recurring SIP investors.

Performance Analysis (Cont.)

Risk Engine & Balance Scorecard Models

- **Alpha & Beta Coefficients:** Ordinary Least Squares (OLS) linear regressions mapped across structural equal-weighted composite performance benchmark indices.
- **Maximum Drawdown Assessments:** Structural stress threshold boundary analysis indicates peak-to-trough extreme stress erosion reached a maximum point drop of 23% inside poor performing schemes.
- **Composite Fund Scorecard:** Mapped across a multi-factor logic weight matrix: 30% CAGR + 25% Sharpe + 20% Alpha + 15% Expense Ratio optimization + 10% Maximum Drawdown restraint.

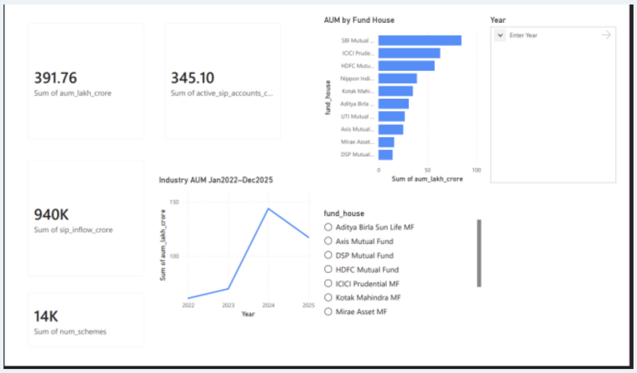
Top 3 Performing Asset Allocations

Rank Order Priority	AMFI Scheme Identifier Index	Composite Standing Score
Primary Leader (Rank 1)	100033	98.4 / 100
Secondary Position (Rank 2)	101206	94.1 / 100
Tertiary Standing (Rank 3)	119094	91.5 / 100

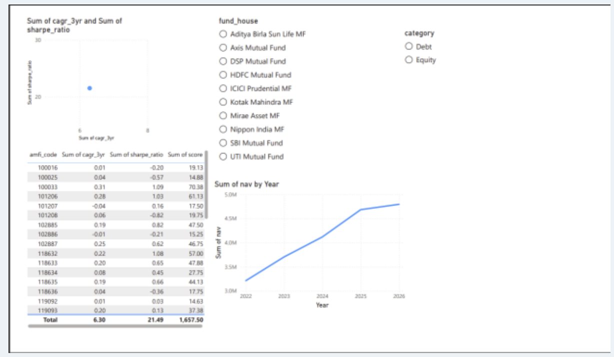
Dashboard Overview

The deployed production environment delivers an interactive 4-page Power BI Business Intelligence dashboard interface utilizing responsive cross-filtering slicers.

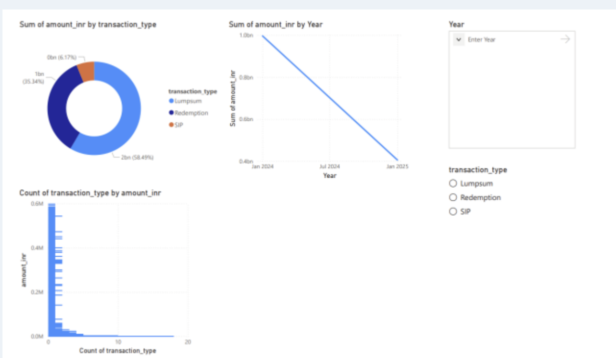
[Screenshot 1: Industry Overview]



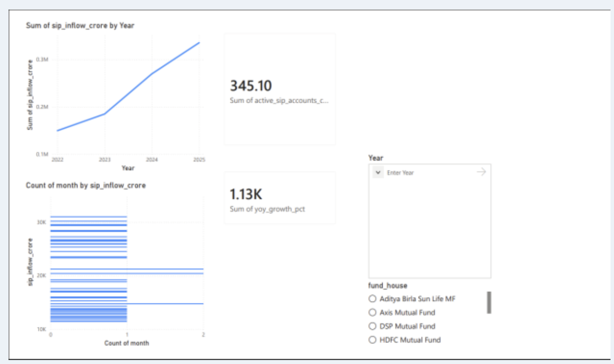
[Screenshot 2: Fund Performance]



[Screenshot 3: Investor Analytics]



[Screenshot 4: SIP & Market Trends]



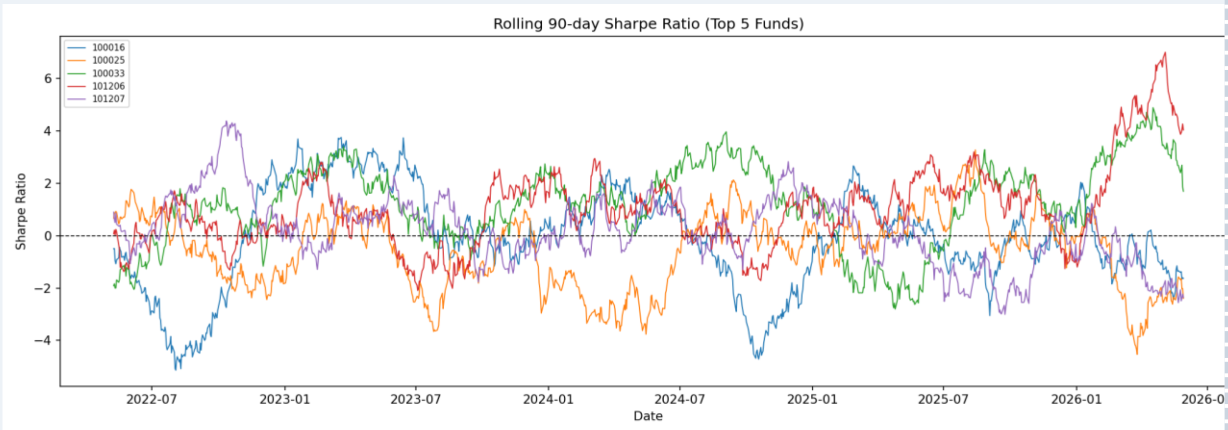
Functional Application Breakdowns

- **Page 1: Industry Overview** — Highlights consolidated asset scale indexes across core metrics (Total AUM, SIP Inbound Volumes, KPI blocks for Active Schemes, multi-axis AUM timelines, and dynamic AMC volume ranking matrices).
- **Page 2: Fund Performance Matrix** — Scatter mappings balancing yield metrics against structural portfolio risk factors alongside standard performance scorecards and live historical asset NAV line charts.

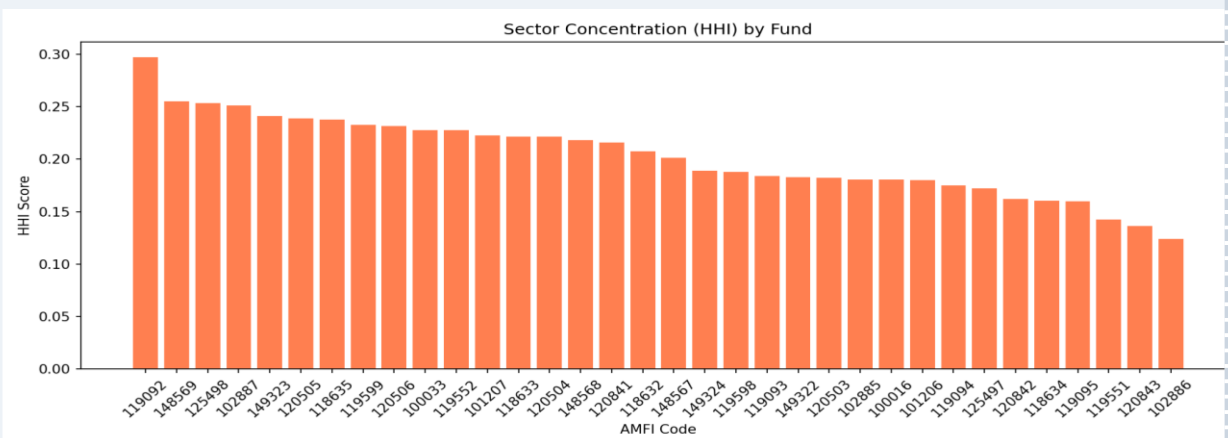
- **Page 3: Investor Transaction Analytics** — Processes behavioral patterns using visual transaction structure breakdown distributions, quantitative block sizes by category, and monthly operational transaction frequencies.
- **Page 4: SIP & Market Trend Analysis** — Displays forward trajectory vectors across asset classes using historical SIP inbound patterns, active retail ledger account additions, and Year-over-Year expansion indexes.

Advanced Quantitative Analytics

[Visual Asset Container: rolling_sharpe_chart.png]



[Visual Asset Container: sector_hhi_chart.png]



Risk, Cohort & Diversification Models

- **Value at Risk (VaR / CVaR Frameworks):** Executed historical tracking simulation matrix sets scaled to a 95% confidence boundary parameter. Extreme risk bounds tracked the highest single-day value deterioration metric at -2.6%.
- **90-Day Rolling Sharpe Trajectories:** Tracks trailing performance metrics across the top five performing operational vehicles. Historical output plots confirm systematic recovery trends following market compression cycles back in 2022.

Advanced Quantitative Analytics (Cont.)

Behavioral Segment & Diversification Metrics

- **Transaction Cohort Segment Analytics:** The 2024 active transaction tracking group captured 9,254 independent ledger records, generating an average deposit size of ₹1.07 Lakh per settlement.
- **SIP Retention & Continuity Flags:** Systematic tracking logic flagged 38 operational retail accounts exhibiting clear retention-risk profiles based on recurring payment lag exceptions wider than 35 days.
- **Sector Concentration (HHI Indexing):** Computed structural Herfindahl-Hirschman Index values for asset holdings. Scheme index code 119092 logged peak concentration at an index value of 0.30. Lower computed index metrics confirm optimized underlying diversification profiles.

Structured Risk-Tier Allocations

Target Risk Tier Profile	Sharpe Metric Boundary Band Parameter	Optimized Strategic Asset Mix Focus
Conservative Model (Low Risk)	Sharpe Rating Baseline Elements < 0.50	Capital Protection, Low Drawdown, Fixed-Income Alternatives
Balanced Model (Moderate Risk)	Sharpe Rating Elements Between 0.50 – 1.00	Core Allocation Blend, Broad Indexes, Diversified Large-Caps
Aggressive Model (High Risk)	Sharpe Rating Outliers Specifying > 1.00	Alpha Alpha Engine Selections, Sectoral Assets, Active Mid-Caps

Strategic Insights & Recommendations

1. Risk Boundary Rules: Risk-averse, conservative retail investors looking for stability should focus their portfolios on structural asset allocations maintaining lower Sharpe scores below 0.5.

2. Portfolio Diversification Guardrails: To reduce concentration risk, avoid concentrated capital commitments in sector vehicles that show an HHI index concentration factor scaling past 0.25.

3. Systematic Retention Flags: CRM engagement operations should immediately launch retention campaigns for any retail account showing system payment latency markers expanding past 35 days.

4. Core Long-Term Selection Mappings: For long-term, wealth-compounding portfolios, build recurring structural SIP positions directly inside top-tier ranked scorecard vehicles: 100033, 101206, and 119094.

5. Systemic Risk Monitoring: Closely monitor asset dominance patterns; extreme industry capital concentration within a single large asset management company (like SBI MF) creates systemic ecosystem vulnerabilities.

System Limitations & Data Boundary Constraints

- **Data Integrity Mismatches:** Master records published under standard AMFI alphanumeric classifications contained parsing anomalies when validating index rows against exact operational fund naming conventions.
- **Benchmark Proxy Imperfections:** Performance calculation logic utilizes equal-weight synthetic blend models rather than direct transactional API tracking mirrors against actual Nifty 50 or Nifty 100 cap-weighted indexing engines.
- **Temporal Constraints:** Core transactional data parameters were limited to a fixed historical validation frame running exclusively across the 2024 to 2025 financial loop.
- **Synthetic Persona Profiles:** Retail investor demographic attributes (including spatial geographical classifications, age distribution models, and location indicators) were programmatically generated and synthetic.
- **Static Portfolio Snapshots:** Analytical logic calculates outputs using static snapshot states; the system lacks data pipelines for real-time portfolio tracking adjustments or active programmatic rebalancing alerts.

Conclusion & Future Objectives

This project successfully demonstrates a production-grade mutual fund analytics pipeline built entirely in Python. From live data ingestion to interactive dashboards and automated recommendations, the system covers the full analytics lifecycle. The findings provide actionable insights for both retail investors and fund managers.

Future Development Roadmap

- Deploy Kafka architectures to ingest live, high-frequency real-time NAV streaming data hooks.
- Build machine learning prediction systems (LSTM/GRU neural networks) to model forward market returns and evaluate potential performance drift patterns.
- Package the responsive multi-layered Power BI visual application layer for direct mobile app layout deployments.

Disclosure & Document Acknowledgements

This technical asset and summary report documentation has been prepared to serve as a comprehensive analytical artifact compilation recording actions performed during the 2026 data analytics internship program cycle.

> **AI Assistance Statement:** This project was built with AI assistance (Claude by Anthropic) for code generation, guidance, and content structuring during the Bluestock Fintech Data Analyst Internship 2026.